

Appendix 2: FKB327-002 protocol outline

CLINICAL PROTOCOL

A Randomised, Blinded, Active-Controlled Study to Compare FKB327 Efficacy and Safety with the Comparator Humira[®] in Rheumatoid Arthritis Patients Inadequately Controlled on Methotrexate (ARABESC)

PROTOCOL SUMMARY

Study Title:

A Randomised, Blinded, Active Controlled Study to Compare FKB327 Efficacy and Safety with the Comparator Humira® in Rheumatoid Arthritis Patients Inadequately Controlled on Methotrexate (ARABESC)

Protocol Number:

FKB327-002

Protocol Acronym:

ARABESC

Name of Investigational Product:

FKB327

Name of Active Ingredient:

Adalimumab

Study Phase:

Phase 3

Number of Patients Planned:

680 patients (approximately 340 patients per arm)

Number of Sites Planned:

Approximately 150 sites in the United States, Canada, European Union, Eastern Europe, and South America

Study Population:

Male and female patients with moderate to severe active rheumatoid arthritis (RA) taking a stable dose of methotrexate (MTX) and who require additional therapy due to an inadequate response to MTX

Primary Objective:

The primary objective of this study is to assess the efficacy of FKB327 compared with Humira®, when each is administered in combination with MTX

Secondary Objectives:

- To compare the safety profiles of FKB327 and Humira, each in combination with MTX treatment
- To assess the efficacy profiles of FKB327 and Humira over time, including initial onset of effect
- To compare the proportions of patients on FKB327 and Humira, who develop anti-drug antibodies (ADAs) and to summarise the distribution of the level of ADA activity between patients on FKB327 and Humira
- To compare the steady state pharmacokinetics (PK) of FKB327 and Humira administered by multiple dosing in patients with RA receiving concomitant treatment with MTX

Key Inclusion Criteria:

1. Male or female aged 18 years or over
2. Patients have been diagnosed with RA for at least 3 months
3. Patient has active RA
4. Patient has taken a stable dose of methotrexate for at least 3 months

Key Exclusion Criteria:

1. Patient has been previously treated with adalimumab
2. Patient has been previously treated or has ongoing treatment with prohibited medications
3. Patient has been immunised with a live or attenuated vaccine in past 4 weeks
4. Patient has positive result for HIV, HBV, HCV or TB infection

Study Drug:

Patients will be administered subcutaneous (sc) FKB327 40 mg every other week (eow) or sc Humira 40 mg eow. Patients will continue to take their usual dose of MTX (10-25 mg/week) throughout the study with concomitant folic/folinic acid at a dose of at least 5 mg/week.

Blinding and Randomisation:

Patients will be randomised to receive either FKB327 40 mg eow or Humira 40 mg eow in a 1:1 ratio

Study Design and Methodology:

This is a Phase 3, multi-centre, randomised, double-blind, parallel arm, active-comparator, equivalence study, during which FKB327 or Humira will be administered by multiple dosing eow for 22 weeks in patients with active RA who are already taking MTX at a stable dose (10 to 25 mg/week) for a minimum of 8 weeks prior to Screening who require additional therapy to control their disease.

Patients will be randomised to receive sc treatment with FKB327 40 mg eow or Humira 40 mg eow from Week 0 to Week 22. Scheduled study visits will be conducted at Screening, Baseline, and at Weeks 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, and 24. At Week 24, eligible patients will, at the discretion of the Investigator, and with the patient's informed consent enter the open-label extension study (Study FKB327-003). Patients not entering the extension study will instead attend the Week 26 follow-up visit.

Primary Efficacy Endpoint:

- The primary efficacy endpoint is the ACR20 response rate at Week 24

Secondary Efficacy Endpoints:

- DAS28-CRP score at Week 24
- ACR20, ACR50 and ACR70 response rates over time
- Values of the individual ACR core set variables over time
- DAS28-CRP score and change in DAS28-CRP score over time
- DAS28-ESR at Weeks 12 and 24

Other Endpoints:

- Proportion of patients developing ADAs
- Trough adalimumab concentration

Safety Endpoints:

- Adverse events (AEs)
- Serious adverse events (SAEs), including serious infections and malignancies
- Vital signs
- Laboratory abnormalities

Assessment**Assessment of Efficacy****Primary Efficacy Variable:**

- ACR20

Secondary Efficacy Variable:

- DAS28
- ACR50 and ACR70 Response Rates

Assessment of Safety

- Clinical Laboratory Tests
- Pharmacodynamic Markers
- Pregnancy Testing
- Physical Examination
- Vital Signs
- Electrocardiograms
- Chest X-ray
- Injection Site Assessment
- Immunogenicity Assessments
- Pharmacokinetic Assessments

Statistical Methods for primary endpoint:

The primary endpoint is the ACR20 response rate at Week 24 analysed using the Full Analysis Set (FAS) and the Per-protocol Analysis Set (PPAS). Two sets of equivalence margins ($\pm 13\%$ using 95% CI and -12% to $+15\%$ using 90% CI) around the difference in ACR20 response rate will be applied in the primary analysis.

SCHEDULE OF EVENTS

Study Schedule

Study Visit	Screening	Baseline	Treatment Period												
Week	≤4 weeks	0	2	4	6	8	10	12	14	16	18	20	22	24	26 (EOS)
Informed consent	•														
Eligibility	•	•													
Medical history, Demographics	•														
Randomisation		•													
Study medication administration (mg)		40	40	40	40	40	40	40	40	40	40	40	40		
Concomitant MTX and folate															
Physical examination, Vital signs, ECG, Chest X-ray, Laboratory testing including CRP, ESR are assessed at the schedule defined by protocol															
Tender/Swollen Joint Counts	•	•	•	•		•		•		•		•		•	
Patient's global assessment	•	•	•	•		•		•		•		•		•	
Patient's pain VAS		•	•	•		•		•		•		•		•	
HAQ-DI		•	•	•		•		•		•		•		•	
Physician's global assessment		•	•	•		•		•		•		•		•	
Serum drug concentration		•	•	•				•				•		•	
ADA test		•	•	•				•						•	
Injection site assessment (including injection site pain VAS)		•													
Adverse events															